

42. Which of the following plants is not helpful in nitrogen fixation?

- (a) Gram
- (b) Pea
- (c) Bean
- (d) Paddy
- (e) None of the above

Chattisgarh P.C.S. (Pre) 2017

Ans. (d)

Among the given options, paddy plant is not helpful in nitrogen fixation.

43. Which of the following is not a cash crop?

- (a) Jute
- (b) Groundnut
- (c) Jowar
- (d) Sugarcane
- (e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (c)

Among the given options, Jowar is not a cash crop, whereas jute, sugarcane and groundnut are cash crops.

44. With reference to the cultivation of Kharif crops in India in the last five years, consider the following statements:

1. Area under rice cultivation is the highest.
2. Area under the cultivation of jowar is more than that of oilseeds.
3. Area of cotton cultivation is more than that of sugarcane.
4. Area under sugarcane cultivation has steadily decreased.

Which of the statement/s given above is/are correct?

- (a) 1 and 3 only
- (b) 2, 3 and 4 only
- (c) 2 and 4 only
- (d) 1, 2, 3 and 4

I.A.S. (Pre) 2019

Ans. (a)

1. The area under rice cultivation in India, in 2014-15 was 44.1 million hectares. In 2023-24 it became 47.8 million hectares which is the highest amongst all.
2. The area under jowar cultivation 4.1 million hectares and oilseed area is 30.2 million hectares in 2023-24. Hence area under oilseed cultivation is more than Jowar cultivation.
3. The area under sugarcane cultivation 5.1 million hectare (2014-15) and 5.7 million hectares (2023-24). Hence it has not a steady decrease for sugarcane.
4. The area under cotton cultivation was 12.8 million hectares (2014-15) & 12.7 million hectares (2023-24). Hence area under cotton cultivation is more than sugar cultivation.

Cash Crops

i. Cotton :

*Cotton is the second major commercial crop of India. It is raw material for cotton textile industry. Cotton belongs to the **Malvaceae Family**. There are two major varieties of cotton found in the world. **(i) Old world cotton (ii) New world cotton**

*Cotton is an indigenous plant of India. It is also mentioned in the ancient text such as Rigveda and Manusmriti. India is the first country in the world where hybrid cotton has been developed resulting in enhanced production. *Black soil is most suitable for its cultivation. The temperature ranging between 25°C-35°C, 210 Frost Free day, clear sky, bright sunshine and annual rainfall of 75-100 cms are favourable conditions for its cultivation.

North-Western and western India are major producing regions of cotton. *According to data from the Economic Survey 2024-25, the three leading states in cotton production are (in 2023-24)-

1. Gujarat (9.06 Million tonnes)
2. Maharashtra (8.05 Million tonnes)
3. Telangana (5.08 Million tonnes)

*Maharashtra is largely covered under black soil which is most suitable soil for cotton cultivation. Cotton is known as White gold in Maharashtra. *Cotton cultivation in Madhya Pradesh is done on the western side of the state e.g. **Plateau of Malwa** and **Narmada Valley**. The main cotton producing district are East and West Nimar, Dhar, Dewas, Shajapur, Ujjain, Ratlam, Khandwa, Hoshangabad and Rajgarh.

*Khandwa - Khargone region is also known as '**white gold**' region due to the cultivation of cotton.

*Cotton Fibers are collected from cotton seeds.

1. "The crop is subtropical in nature. A hard frost is injurious to it. It requires at least 210 frost-free days and 50 to 100 centimeters of rainfall for its growth. A light well-drained soil capable of retaining moisture is ideally suited for the cultivation of the crop." Which one of the following is that crop?

- (a) Cotton
- (b) Jute
- (c) Sugarcane
- (d) Tea

I.A.S. (Pre) 2020

Ans. (a)

Cotton : This subtropical crop is suitable for drier parts of the black cotton soil of the Deccan plateau. It requires high temperature, light rainfall (50-100 cm), 210 frost-free days and bright sunshine for its growth.

2. Consider the following statements with reference to cotton crops:

1. Cotton is mostly a Kharif crop.
2. Cotton crop requires moderate amounts of rainfall varying between 50-75 cm annually.

Which of the above statements is/are correct?

- (a) Both 1 and 2 (b) Neither 1 nor 2
(c) Only 1 (d) Only 2

UPPSC (Pre) 2024

Ans. (a)

Cotton is primarily a Kharif season cash crop. The most suitable temperature for Cotton Production ranges between 21°C and 27°C. It cannot tolerate temperatures below 15°C or above 43°C. Frost is harmful of cotton and it thrives in semi-arid conditions. Cotton is a tropical and subtropical crop. Generally it requires moderate rainfall of 70 to 120 cm. (according to R.C. Tiwari, 50-75 cm. of rainfall) for its production.

3. Which one of the following is a group of millet crops?

- (a) Bajra, Maize, Kodo, Sorghum
(b) Ragi, Bajra, Kodo, Moong
(c) Kodo, Bajra, Maize, Kangani
(d) Sorghum, Kodo, Kangani, Ragi
(e) Question not attempted

Raj. P.C.S. (Pre) 2023

Ans. (d)

Sorghum, Kodo, Kangani and Ragi come under millet crops. Notably, Indian millets are a group of nutritiously rich, drought tolerant crops mostly grown in the arid and semi-arid regions of India. Significantly, millet is a type of grain that is popular in many parts of the world, especially in Africa and Asia.

4. Which one of the following areas of India produces the largest amount of cotton?

- (a) North Western India and Gangetic West Bengal
(b) North Western and Western India
(c) Western and Southern India
(d) Plains of Northern India

I.A.S. (Pre) 1996

Ans. (b)

When the question was asked North western and western India were the major producers of cotton. According to data from the Economic Survey 2024-2025, in 2023-24 the cotton cultivation in respect of Major crop is done on 12.7 million hectare area.

5. The largest producer of cotton in India is –

- (a) Maharashtra (b) Gujarat
(c) Punjab (d) Haryana

44th B.P.S.C. (Pre) 2000

Ans. (b)

According to the data of 2023-24, the three largest cotton producing states in India are 1. Gujarat, 2. Maharashtra, 3. Telangana.

6. Which districts are Known as areas of 'White Gold' because of cotton-cultivation.

- (a) Ratlam-Khandwa (b) Khandwa-Khargone
(c) Ujjain-Shajapur (d) Dhar-Jhabua

M.P.P.C.S. (Pre) 1997

Ans. (b)

Khandwa-Khargone are known as the areas of 'White Gold' because of cotton cultivation.

7. Which one of the following crops is known as "White Gold" in Maharashtra.

- (a) Wheat (b) Maize
(c) Cotton (d) Sugar Cane

U.P. Lower Sub. (Pre) 2002

Ans. (c)

Black soil is suitable for cotton cultivation which covers the maximum area in Maharashtra. Cotton crop is known as White Gold in Maharashtra.

8. Consider the following statements:

1. India is the original home of the cotton plant
2. India is the first country in the world to develop hybrid cotton variety leading to increased production

Which of these statements is/are correct?

- (a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

I.A.S. (Pre) 2003

Ans. (c)

India is the original home of cotton plant. It also finds mention in Rigveda and Manusmriti. India is also the first country to develop a hybrid variety of cotton which has resulted in increased production.

9. Cotton fibers are obtained from –

- (a) Leaves (b) Seed
(c) Stem (d) Root

U.P.P.C.S. (Mains) 2009

Ans. (b)

Cotton fibers are obtained from seeds.

10. Cotton is facing competition from sugarcane in the region of the black soil in Maharashtra. This is due to-

- (a) Declined yield of cotton in this region
- (b) A general change in the climate of this region
- (c) Expansion of irrigation has made this region suitable for sugarcane cultivation which is a more profitable crop
- (d) Increasing demand for sugar and the rising price of sugar in the country.

Jharkhand P.C.S. (Pre) 2011

Ans. (c)

Cotton is facing competition from sugarcane in the region of black soil in Maharashtra due to the expansion of irrigation. This region has become more suitable for sugarcane cultivation. Sugarcane cultivation is more profitable than the cotton crop.

11. The average yield of cotton is highest in India for the state of –

- (a) West Bengal
- (b) Bihar
- (c) Assam
- (d) Andhra Pradesh
- (e) None of these/More than one of these

63rd B.P.S.C. (Pre) 2017

Ans. (d)

When the question was asked option (d) was correct answer. According to Agricultural Statistics at a glance, 2023 Gujarat (2022-23) has the highest productivity of cotton.

12. Which Indian state has the largest number of Cotton Textile Mills?

- (a) Madhya Pradesh
- (b) Maharashtra
- (c) Gujarat
- (d) West Bengal
- (e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (e)

Maharashtra is the state having the largest number of cotton Textile mills among the given options.

ii. Sugarcane :

*It is a tropical as well as a sub-tropical crop. It grows well in a hot and humid climate with the temperature ranging between 21°C to 27°C and an annual rainfall between 75 cm and 100 cm. Irrigation is required in the regions of low rainfall.

*According to the data of the year 2023-24 three leading states in Sugarcane production are -

- 1. Uttar Pradesh (215.81 Million tonnes).
- 2. Maharashtra (112.09 Million Tonnes)

3. Karnataka (41.81 Million tonnes)

The sustainable sugarcane initiative (SSI) is an innovative set of agronomic practices to increase cane yields significantly. This system was initiated in the year 2009 for improvement in sugarcane agriculture. The initiation of this method was a joint venture of ICRISAT and WWF.

*The breeding of Sugarcane in India is being done in **Coimbatore. Sugarcane breeding Institute** was established in 1912 in Coimbatore (Tamil Nadu). On 1 April 1969, it was made a part of ICAR.

1. Among the following, which one is the least water-efficient crop?

- (a) sugarcane
- (b) Sunflower
- (c) Pearl millet
- (d) Red gram

I.A.S. (Pre) 2021

Ans. (a)

The least water efficient crop among the given option is sugarcane, i.e., maximum water is required to grow sugarcane.

2. Which Indian state has the largest area under sugarcane cultivation?

- (a) Maharashtra
- (b) Uttar Pradesh
- (c) Andhra Pradesh
- (d) Madhya Pradesh

U.P.P.C.S. (Pre) 2015

Ans. (b)

According to data for the year 2022-23 the largest area under sugarcane cultivation is in Uttar Pradesh (2.79 Million Hectares) which is followed by Maharashtra (1.36 Million Hectares).

3. Which of the following crops has the largest percentage of irrigated area of its net sown area in India?

- (a) Wheat
- (b) Rice
- (c) Oilseed
- (d) Sugarcane

U.P.U.D.A./L.D.A. (Pre) 2013

Ans. (d)

When the question was asked option (d) was correct. The percentage of area under irrigation (2021-22) of the given crops is as follows –

(Crop)	Area Under Irrigated (Percentage)
Sugarcane	98.73
Wheat	95.29
Rice	67.56
Oilseed	37.51

4. The correct sequence in decreasing order of the four sugarcane producing states in India is:

- (a) Maharashtra, U.P., Tamil Nadu, Andhra Pradesh
- (b) U.P., Maharashtra, Tamil Nadu, Andhra Pradesh
- (c) Maharashtra, U.P., Andhra Pradesh, Tamil Nadu
- (d) U.P., Maharashtra, Andhra Pradesh, Tamil Nadu.

I.A.S. (Pre) 2000
44th B.P.S.C. (Pre) 2000

Ans. (b)

According to the data available at the time when this question was asked, the four sugarcane producing states in descending order of their sugarcane production were as follows –

- (1) Uttar Pradesh
- (2) Maharashtra
- (3) Tamil Nadu
- (4) Andhra Pradesh

According to Statistics at a Glance, 2023 in 2022-23 data, the four largest sugarcane producing states are –

- (1) Uttar Pradesh
- (2) Maharashtra
- (3) Karnataka
- (4) Tamil Nadu
- (5) Gujarat

5. The leading sugarcane producing states in India is

- (a) Punjab
- (b) Andhra Pradesh
- (c) Maharashtra
- (d) Uttar Pradesh

U.P.P.C.S. (Mains) 2013

Ans. (d)

See the explanation of the above question.

6. Which States in India are the largest producers of sugarcane?

- (a) Bihar and Uttar Pradesh
- (b) Uttar Pradesh and Rajasthan
- (c) Andhra Pradesh and Jammu and Kashmir
- (d) Punjab and Himachal Pradesh

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (a)

When the question was asked option (a) was the correct answer. According to data of the year 2022-23, the eight largest sugarcane producing states are – Uttar Pradesh, Maharashtra, Karnataka, Tamil Nadu, Gujarat, Bihar, Haryana and Punjab.

7. What is the correct sequence of the states in descending order of production of sugarcane in 2013 ?

- (a) Karnataka, Maharashtra, Haryana, U.P.
- (b) U.P., Maharashtra, Karnataka, Haryana

- (c) Haryana, Karnataka, Maharashtra, U.P.
- (d) U.P., Karnataka, Maharashtra, Haryana

U.P.P.C.S. (Pre) 2014

Ans. (b)

See the explanation of the above question.

8. Consider the following statements:

1. Molasses is a by-product of the sugar production process.
2. Bagasse obtained from sugar mills is used as a fuel in the boilers to generate steam in sugar factories.
3. Sugar can only be produced from sugarcane as the raw material.

Which of these statements is/are correct?

- (a) 1 and 2
- (b) 2 and 3
- (c) 1 and 3
- (d) 1, 2 and 3

I.A.S. (Pre) 2003

Ans. (a)

Molasses is the main by-product of sugar production process which is used for alcohol production. Bagasse is used as a fuel to generate steam in sugar mills. Sugar can not only be produced from sugarcane but also from beetroot.

9. What are the significances of a practical approach to sugarcane production known as 'Sustainable Sugarcane Initiative'?

1. Seed cost is very low in this compared to the conventional method of cultivation.
2. Drip irrigation can be practised very effectively in this.
3. There is no application of chemical/inorganic fertilizers at all in this.
4. The scope for intercropping is more in this compared to the conventional method of cultivation.

Select the correct answer using the code given below –

- (a) 1 and 3 only
- (b) 1, 2 and 4 only
- (c) 2, 3 and 4 only
- (d) 1, 2, 3 and 4

I.A.S. (Pre) 2014

Ans. (b)

The Sustainable Sugarcane Initiative (SSI) is an innovative set of agronomic practices that involves using fewer seeds, raising seedlings in a nursery, and following new planting methods with wider plant spacing, and better water and nutrient management to increase the cane yields significantly. This initiative was started by the combined efforts of WWF-ICRISAT in 2009. Hence, statement (3) is false.

10. Sucrose content in sugarcane decreases

- (a) If high rainfall occurs during the period of growth of the plant
- (b) If frost occurs during the period of ripening

- (c) If there is fluctuation in temperature during the period of growth of the plant
(d) if there is high temperature during the time of ripening

I.A.S. (Pre) 1993

Ans. (b)

Sugarcane is a tropical plant. The areas having a temperature of 21° to 27°C and an average rainfall of 75-100 cm are suitable for its cultivation. Thus, if frost occurs during the period of ripening, the Sucrose content in the sugarcane decreases.

11. The first three leading producers of sugar in India are:

- (a) Maharashtra, U.P., Tamil Nadu
(b) U.P., Maharashtra, Bihar
(c) U.P., Maharashtra, Karnataka
(d) Bihar, U.P., Karnataka, Maharashtra

U.P.P.C.S. (Pre) 1999

U.P.P.C.S. (Pre) 1996

Ans. (c)

Top three sugar producing States of India (at the time when this question was asked) were 1. Uttar Pradesh, 2. Maharashtra and 3. Karnataka. According to the data of Indian Sugar Mills Association (as on April 15, 2024), Maharashtra is at the top in sugar production while Uttar Pradesh is in second place.

12. Which of the following represents the correct sequence in descending order, of the four largest sugar producing States of India in 2015-16?

- (a) Maharashtra, U.P., Karnataka, Tamil Nadu
(b) U.P., Maharashtra, Karnataka, Tamil Nadu
(c) Maharashtra, U.P., Tamil Nadu, Karnataka
(d) U.P., Karnataka, Maharashtra, Tamil Nadu

U.P. P.C.S. (Mains) 2017

Ans. (a)

When the question was asked option (a) was correct answer. According to the data from the Indian Sugar Mills Association, Maharashtra is at the top in sugar production while Uttar Pradesh is at the second place.

13. Which of the following statements is/ are correct regarding sugar Industry?

1. India accounts for more than 15% of the total sugar production of the world.
2. Sugar Industry is the second largest Agricultural base industry in India.
3. India is the biggest consumer of sugar.
4. India is the largest producer of sugar.

Code :

- (c) 1 and 2 (b) 3 and 4

(c) 1, 2 and 3

(d) 1 and 4

U.P.P.C.S. (Mains) 2009

Ans. (c)

As per global data (2023/2024) issued by U.S. Department of Agricultural, the world's total production of sugar was 183.83 million Metric tonnes of which Brazil produced 45.54 million tonnes sugar while India's share was 34.00 million tonnes out of the total sugar production. Sugar Industry is the second largest agriculture-based industry in India after cotton. When the question was asked India accounts for 15.8% of the world's sugar production. Thus, statement (1) and (2) are correct. Brazil is the largest producer of sugar not India. Thus, statement (4) is incorrect. India is the largest consumer of sugar. Thus, statement (3) is also correct.

14. Sakkar Nagar is an important sugar producing centre of

- (a) Haryana (b) Andhra Pradesh
(c) Maharashtra (d) Uttar Pradesh

U.P.P.C.S. (Spl) (Mains) 2008

Ans. (b)

Sakkarnagar is an important sugar producing centre located in Andhra Pradesh (now in Telangana).

15. Which state is called the 'Sugar Bowl' of India ?

- (a) Andhra Pradesh (b) Bihar
(c) Punjab (d) Uttar Pradesh

U.P.P.C.S. (Spl) (Mains) 2008

Ans. (d)

Uttar Pradesh is known as the 'Sugar Bowl' of India.

16. Which one among the following States of India is called 'Sugar Bowl'?

- (a) Uttar Pradesh (b) Maharashtra
(c) Bihar (d) Haryana
(e) None of the above/More than one of the above.

64th B.P.S.C. (Pre) 2018

Ans. (a)

Uttar Pradesh is called the sugar bowl of India. Presently (as on April 15, 2024) Uttar Pradesh. It is the second largest producer of sugar in India.

17. The first Sugar Mill in India was set up in 1903 at

- (a) Pratapgarh (b) Pratappur
(c) Mawana (d) Balrampur

U.P.P.C.S. (Mains) 2013

Ans. (b)

The first Sugar Mill was set up in India in 1903 at Pratappur in Deoria district of Uttar Pradesh.

18. Which factors have facilitated the locational shift of the sugar industry from North India to South India ?

1. Per acre higher yield of sugarcane
2. Higher sucrose content of sugar cane
3. Longer crushing season
4. Cheap labour

Select the correct answer using the code given below—

- (a) 1, and 2 (b) 2 and 3
(c) 1, 2 and 3 (d) 2, 3 and 4

U.P.P.C.S. (Mains) 2007

Ans. (c)

The factors that facilitated the locational shift of the sugar industry from North India to South India are per acre higher yield of sugarcane, higher sucrose content of sugarcane and longer crushing season.

19. Breeding work on sugarcane is being done at –

- (a) Lucknow (b) Shahjahanpur
(c) Shravasti (d) Coimbatore

U.P.P.C.S. (Mains) 2009

Ans. (d)

Breeding work on sugarcane is being done in Coimbatore, India. **Sugarcane Breeding Institute** is a central research institute established in Coimbatore (Tamil Nadu), India. It was established in 1912 under Indian Council of Agricultural Research.

20. Sugarcane seeds are produced in –

- (a) I.I.S.R , Lucknow
(b) IARI, New Delhi
(c) Directorate of Seed Research, Mau
(d) SBI, Coimbatore

U.P.U.D.A./L.D.A (Spl.) (Mains), 2010

Ans. (d)

Sugarcane Breeding Institute is a Central Research Institute in Coimbatore, India. It was established in 1912 under Indian Council of Agricultural Research. It was established to promote research efforts in sugarcane production and is the only sugarcane research institute in the country.

21. What time is taken by Adsali Sugarcane crop to get ripen?

- (a) 6 months (b) 12 months
(c) 18 months (d) 24 months

U.P.U.D.A./L.D.A . (Spl) (Mains) 2010

Ans. (c)

Adsali Sugarcane planting is mainly done in low rainfall areas of Maharashtra. Planting is done in the months of July-August and it takes 16-18 months to get ripened.

22. Consider the statements :

Assertion (A) : Sugar Industry is more developed in South India.

Reason (R) : The per hectare yield of sugarcane and the quantity of sugarcane juice are more in the southern states.

Code :

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

U.P. R.O./A.R.O. (Mains), 2017

Ans. (a)

The sugar industry is more developed in South India. Maharashtra, Karnataka and Tamil Nadu are the major sugar producing states in India. In fact due to favourable climate and black soil area in South India alongwith the ability for moisture conservation and development of tubewell irrigation, South India has high productivity of sugarcane is per hectare yield of sugarcane and high amount of juice in sugarcane. Hence both Assertion (A) and Reason (R) are correct, and Reason (R) is the correct explanation of Assertion (A).

23. Given below are two statements, one is labeled as Assertion (A) and the other as Reason (R).

Assertion (A) : Sugarcane and sugar production in U.P. is more than Maharashtra, but productivity is less.

Reason (R) : Most of the sugar factories in Maharashtra are in the cooperative sector.

Code :

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

U.P.P.C.S. (Pre) 2019

Ans. (b)

According to Indian Sugar Mills Association data, sugarcane and sugar production in U.P. is more than Maharashtra but productivity is less. Assertion (A) is the correct statement. Most of the sugar factories in Maharashtra being in the cooperative sector is also correct. Reason (R) is the correct statement. Reason (R) is not the correct explanation of Assertion (A), but both (A) and (R) are true. So, the correct answer is option (b).